

JEE Main Level Practice Test-7

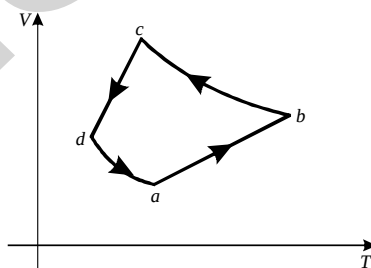
for JEE & NEET Aspirants

**Topic : THERMODYNAMICS &
HEAT TRANSFER**

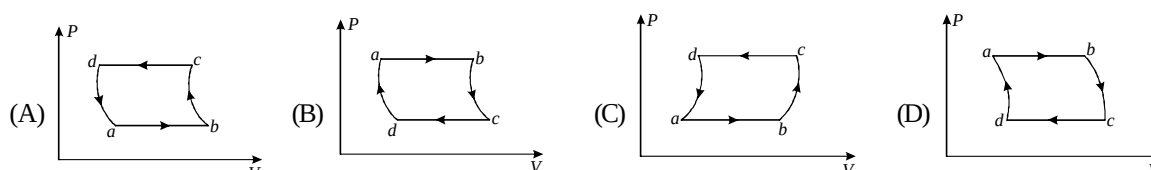
Time: 75Min Marking: +4 -1

Section - A : MCQs with Single Option Correct

- A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127°C and rejects 600 J of heat energy during each cycle. The efficiency of engine and temperature of sink will be :
 (A) 20% and -43°C (B) 40% and -33°C (C) 50% and -20°C (D) 70% and -10°C
- Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q , where V is the volume of the gas. The value of q is $\left(\gamma = \frac{C_p}{C_v}\right)$:
 (A) $\frac{3\gamma+5}{6}$ (B) $\frac{3\gamma-5}{6}$ (C) $\frac{\gamma+1}{2}$ (D) $\frac{\gamma-1}{2}$
- Consider a spherical shell of radius R at temperature T . The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is :
 (A) $T \propto \frac{1}{R}$ (B) $T \propto \frac{1}{R^3}$ (C) $T \propto e^{-R}$ (D) $T \propto e^{-3R}$
- A solid body of constant heat capacity $1 \text{ J/}^\circ\text{C}$ is being heated by keeping it in contact with reservoirs in two ways :
 (i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplied same amount of heat
 (ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat
 In both the cases body is brought from initial temperature 100°C to final temperature 200°C . Entropy change of the body in the two cases respectively is :
 (A) $\ln 2, 2 \ln 2$ (B) $2 \ln 2, 8 \ln 2$ (C) $\ln 2, 4 \ln 2$ (D) $\ln 2, \ln 2$
- An ideal gas goes through a reversible cycle $a \rightarrow b \rightarrow c \rightarrow d$ has the V - T diagram shown below. Process $d \rightarrow a$ and $b \rightarrow c$ are adiabatic.



The corresponding P - V diagram for the process is (all figures are schematic and not drawn to scale) :



6. An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by $P V^n = \text{constant}$, then n is given by (Here C_p and C_v are molar specific heat at constant pressure and constant volume, respectively) :

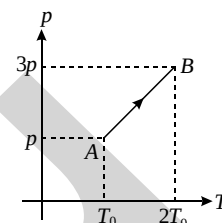
(A) $n = \frac{C_p}{C_v}$ (B) $n = \frac{C - C_p}{C - C_v}$ (C) $n = \frac{C_p - C}{C - C_v}$ (D) $n = \frac{C - C_v}{C - C_p}$

7. If one mole of a monoatomic gas ($\gamma = 5/3$) is mixed with one mole of a diatomic gas ($\gamma = 7/5$), the value of γ for the mixture is :

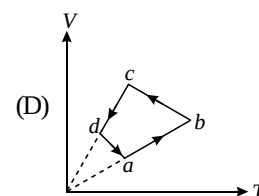
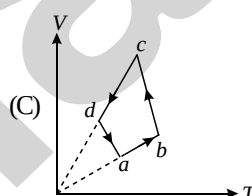
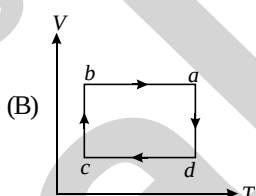
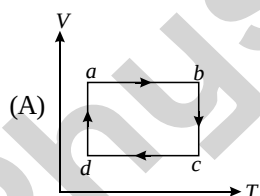
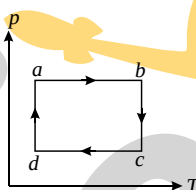
(A) 1.40 (B) 1.50 (C) 1.53 (D) 3.07

8. Pressure versus temperature graph of an ideal gas is shown. The density of gas at point A is ρ_0 , then density of gas at point B would be :

(A) $\frac{3}{4} \rho_0$ (B) $\frac{3}{2} \rho_0$
(C) $\frac{4}{3} \rho_0$ (D) $2\rho_0$



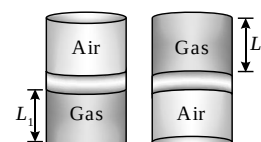
9. A cyclic process is shown on p - T diagram in figure. Which of the curve(s) show(s) the same process on V - T diagram ?



10. An ideal gas is trapped inside a test tube of cross-sectional area $20 \times 10^{-6} \text{ m}^2$ as shown in the figure. The gas occupies a column of length L_1 at the bottom of the tube and is separated from air at atmospheric pressure by a mercury column of mass 0.002 kg. If the tube is quickly turned isothermally, upside down so that mercury column encloses the gas from below. The gas now occupies (a column of length L_2) in the tube. The ratio $\frac{L_2}{L_1}$ is :

[Take, atmospheric pressure = 10^5 Nm^{-2}]

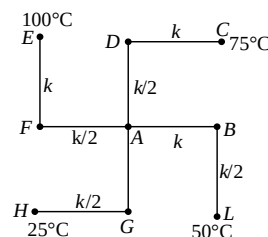
(A) $\frac{102}{101}$ (B) $\frac{101}{99}$
(C) $\frac{99}{100}$ (D) $\frac{100}{99}$



11. At a specific temperature, the energy densities for a black body for three different wavelengths are 10, 19 and 7 units. For these wavelengths if the absorption coefficient of a body is respectively 0.8, 0.3 and 0.9 then the emissive powers of this body for these wavelengths are in the ratio :

(A) 10 : 19 : 7 (B) 8 : 3 : 9 (C) 8 : 5.7 : 6.3 (D) 8 : 9.5 : 4.3

12. All rods have identical length and cross-sectional area. Find the temperature at junction A. The thermal conductivity of rod-AF, rod-GH rod-BL & rod-AD are $k/2$ and the thermal conductivity of rod-EF, rod-CD, rod-AB & rod-AG are k :



- (A) $\frac{125^\circ\text{C}}{4}$ (B) $\frac{125^\circ\text{C}}{2}$
(C) Zero (D) 125°C

13. Two identical blocks of metal are at 20°C and 80°C respectively. The specific heat of the material of the two blocks increases with temperature. Which of the following is true about the final temperature T_f when the two blocks are brought into contact? (Assuming that no heat is lost to the surroundings)

- (A) T_f will be 50°C (B) T_f will be more than 50°C
(C) T_f will be less than 50°C
(D) T_f can be either more than or less than 50°C depending on the precise variation of the specific heat with temperature

14. In an Air conditioner compartment of a train if a material of thermal conductivity k and thickness 10 cm is required to fulfill the purpose. To do so there is a need of two glass each of thickness 1 cm with air gap of 8 cm between them. If thermal conductivity of glass and air are 1 W/mK and 0.01 W/mK then the value of k is :

- (A) 0.0125 W/mK (B) 0.05 W/mK (C) 0.255 W/mK (D) 0.33 W/mK

15. A sphere and a cube of same material, same volume are heated to the same temperature and these are allowed to cool in the same surroundings. The ratio of their initial rate of cooling is :

- (A) $(6\pi)^{1/3}$ (B) $\left(\frac{\pi}{6}\right)^{1/3}$ (C) $\left(\frac{\pi}{3}\right)^{1/3}$ (D) $\left(\frac{2\pi}{3}\right)^{1/3}$

16. Infrared radiations are detected by :

- (A) Spectrometer (B) Pyrometer (C) Nanometer (D) Photometer

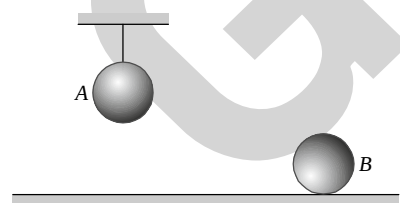
17. Which of the following is more close to a black body :

- (A) Black board paint (B) Green leaves (C) Black marbles (D) Red roses

18. The earth radiates in the infrared region of the spectrum. The spectrum is correctly given by :

- (A) Rayleigh Jeans law (B) Plank's law of radiation (C) Stefan's law of radiation (D) Wien's law

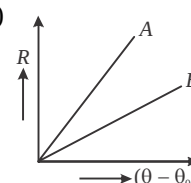
19. Two identical balls, A and B of uniform composition and initially at the same temperature, each absorb exactly the same amount of heat. A is hanging down from the ceiling while B rests on the horizontal floor of the same room. Assuming no subsequent heat loss by the balls which of the following statements is correct about final temperatures T_A and T_B once the balls have reached their final dimension :



- (A) $T_A < T_B$ (B) $T_A = T_B$ (C) $T_A > T_B$ (D) $T_A \leq T_B$

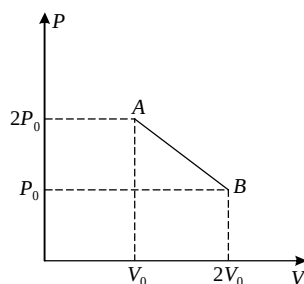
20. Two circular discs A and B with equal radii are blackened. They are heated to same temperature and cooled under identical conditions. What inference do you draw from their cooling curves ? (R is rate of cooling)

- (A) A and B have same specific heats
(B) Specific heat of A is less
(C) Specific heat of B is less
(D) Nothing can be said



Section- B: INTEGER Answer Type Questions

21. 3 moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. The maximum temperature of the gas during the process is $\frac{3P_0V_0}{\lambda R}$. The value of λ is.



22. Horizontal cylindrical vessel is divided into two parts with a thin metal piston. One part of the vessel contains oxygen of mass m , the other the hydrogen of mass $m/4$. What is the equilibrium position of the piston. Calculate the length (in cm) of the part of cylinder with oxygen in it, if the length of the vessel $l = 10$ cm?
23. An ideal monatomic gas initially occupies volume V at pressure P . Now the gas undergoes a process in which pressure of the gas is proportional to its volume. It is found that the rms speed of the gas molecules gets doubled after the process. Heat supplied to the gas in the above said process is xPV . Find x .
24. The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is $n/5$. Find n .
25. A Carnot's engine works as a refrigerator between 250 K and 300 K. If it receives 750 calories of heat from the reservoir at the lower temperature, calculate the amount of heat (calories) rejected at the higher temperature.
26. An ideal gas is expanding such that, pT^3 is a constant. The coefficient of volumetric expansion of the gas is n/T . Find n .
27. Two identical metallic sheets of area $\frac{10^3}{17 \times 27}$ m² are arranged parallel with some separation between them in vacuum. Thermal energy at a constant rate ' P ' is generated in one of the sheets by passing current through it. In steady state, the temperature of the other sheet is found to be 300 K. Calculate the value of P (in kW). Take $\sigma = \frac{17}{3} \times 10^{-8}$ W/m²K⁴.
28. Two spheres of the same material have radii 1 m and 4 m and at temperatures 4000 K and 2000 K respectively. Calculate the ratio of the energy radiated per second by the first sphere to that by the second.
29. 600 J of heat is added to a monoatomic gas on a process in which the gas performs a work of 150 J. The molar heat capacity for the process is KR . Then find K .
30. According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surroundings. Calculate the value of n .

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Section - A : MCQs with Single Option Correct

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|---------|---------|---------|---------|
| 1. (B) | 2. (C) | 3. (A) | 4. (D) |
| 5. (B) | 6. (B) | 7. (B) | 8. (B) |
| 9. (C) | 10. (B) | 11. (C) | 12. (B) |
| 13. (B) | 14. (A) | 15. (B) | 16. (B) |
| 17. (A) | 18. (D) | 19. (C) | 20. (B) |

Section- B: INTEGER Answer Type Questions

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| 21. [4] | 22. [2] | 23. [6] | 24. [2] |
| 25. [900] | 26. [4] | 27. [3] | 28. [1] |
| 29. [2] | 30. [1] | | |

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